

WS 27_Missing Answers

 21. Based on the p -value you can conclude that:

- A. the data provide significant evidence that the proportion of Pakistani adults who oppose coeducation is greater than 50%.
- B. the data provide significant evidence that the majority of Pakistani adults oppose coeducation.
- C. the data do not provide enough evidence to conclude that the proportion of Pakistani adults who oppose coeducation is greater than 50%.
- D. the data provide evidence that H_a is more likely than H_0 (i.e. more than 50% of Pakistani adults oppose coeducation).

B. Recall that our conclusion is based on how small the p -value is. In this case, the p -value is 0.002, which is less than our cutoff of 0.05. This means that if the p -value is less than 0.05, we would reject the null hypothesis (H_0) in favor of the alternative hypothesis (H_a).

The p -value in this case is 0.002, which is less than our cutoff of 0.05. This means that if the p -value is less than 0.05, we would reject the null hypothesis (H_0) in favor of the alternative hypothesis (H_a). Can you now say what this means in context?

Because of the small p -value, we reject the null hypothesis (opposite of the alternative hypothesis) in favor of the alternative hypothesis (opposition is greater than 50%). Thus, we conclude that the majority of Pakistani adults oppose coeducation.

Indeed, since the p -value is small (less than 0.05) we have enough evidence to reject the null hypothesis (H_0) in favor of the alternative hypothesis (H_a). In other words, to conclude that the majority of Pakistani adults oppose coeducation.

 23. When would you conclude that the data provide enough evidence to conclude that the proportion of individuals who oppose coeducation is 50%?

- A. when the p -value is small (less than 0.05).
- B. when the p -value is not small (above 0.05).
- C. when exactly half the individuals in the sample oppose coeducation.
- D. never.

D.

Recall that our conclusion is based on how small the p -value is. H_0 means that if the p -value is less than 0.05, we reject the null hypothesis.

The question here asks when we would accept the null hypothesis (i.e., when we would conclude that the data provide enough evidence to conclude that the proportion of individuals who oppose coeducation is 50%). Under what circumstance would we accept the null hypothesis?

We *never* accept the null hypothesis. The data may not provide enough evidence to reject the null hypothesis, but that does not mean that we have evidence to accept the null.

We never conclude that the data provide enough information to conclude that the proportion of individuals who oppose coeducation is 50% (i.e., we never conclude that the data provide enough information to accept the null hypothesis).

24. An NBC News/*Wall Street Journal* poll conducted in 2011 found that 53 percent of Americans do not support the Tea Party movement. In a poll of 1,000 Americans, 54 percent of the respondents support the Tea Party movement. Has opposition to the Tea Party movement increased?

We tested the following hypotheses at the 5 percent level of significance:

- H_0 : The proportion of Americans this year who oppose the Tea Party movement is 0.53.
- H_a : The proportion of Americans this year who oppose the Tea Party movement is greater than 0.53.

The p -value is 0.026, so we reject the null hypothesis, H_0 , and accept the alternative hypothesis, H_a , that public opposition to the Tea Party movement is greater than 53 percent.

What type of error is possible here?

- A. Type I.
- B. Type II.
- C. neither.
- D. both.

A.

Remember, you are trying to determine what type of error (type I or type II) is possible when making their conclusions.

A type I error is possible only if we reject the H_0 . A type II error is possible only if we fail to reject the H_0 .

We rejected the null hypothesis. With a type I error, we reject a null hypothesis that is true.



26. Suppose that at the end of the five-year study described above, the hormone group have breast cancer and heart disease. This observed difference was so alarmed by the results that the experiment is ended early to protect the health of the participants.

participating in the hormone group. Since the null hypothesis was rejected, this is a type I error.

Given the type of error made in this situation, what could researchers have done to reduce the risk of making a type I error?

They could choose a 0.01 significance level instead of a 0.05 significance level, reducing the risk of making a type I error from 5 percent to 1 percent.